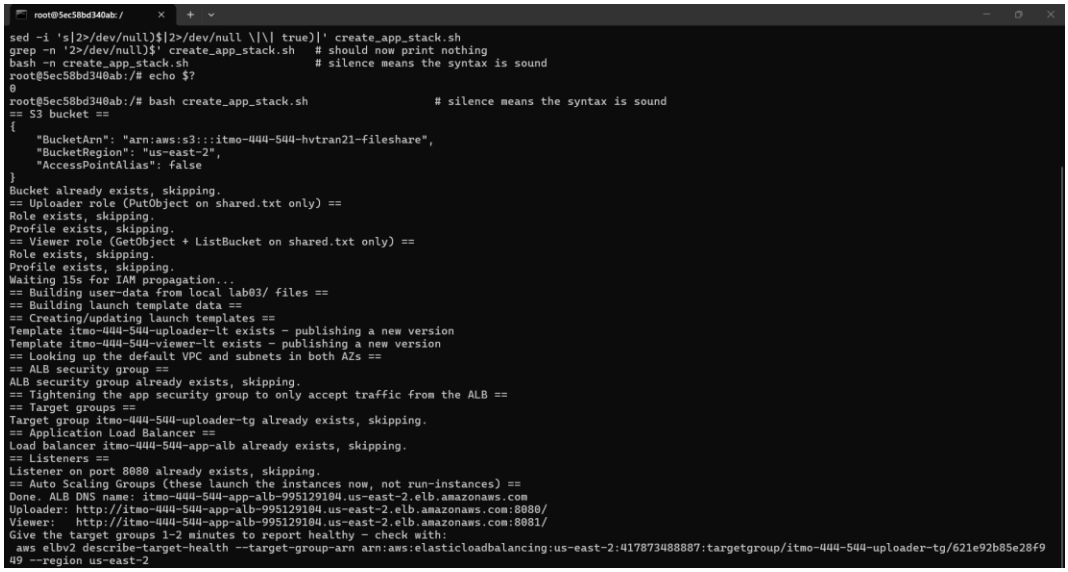
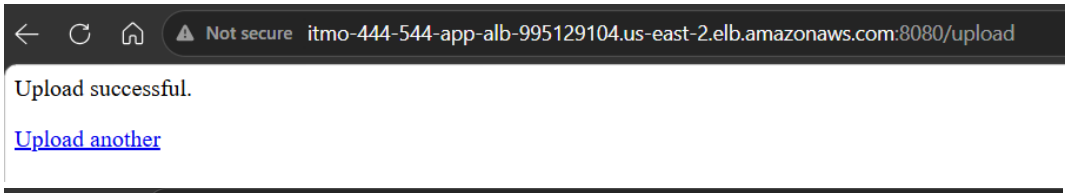
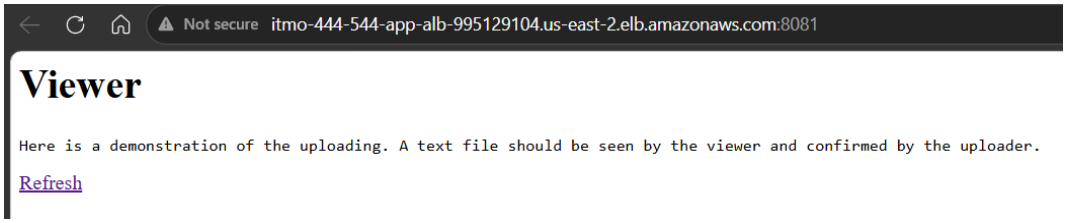
1. Screenshot of create_app_stack.sh completing, showing both http://<alb-dns>:port/ URLs.
   a. 

2. Screenshot of both apps working through the ALB (uploader upload confirmation, viewer showing content).
   a. 

   b. 

3. Screenshot of aws elbv2 describe-target-health showing both target groups healthy.
   a. 

b.

```
aws elbv2 describe-target-health --target-group-arn "$VIEWER_TG_ARN" --region "$AWS_REGION"
{
    "TargetHealthDescriptions": [
        {
            "Target": {
                "Id": "i-041588993091f9e0e",
                "Port": 3000
            },
            "HealthCheckPort": "3000",
            "TargetHealth": {
                "State": "healthy"
            },
            "AdministrativeOverride": {
                "State": "no_override",
                "Reason": "AdministrativeOverride.NoOverride",
                "Description": "No override is currently active on target"
            }
        }
    ]
}
root@5ec58bd340ab:/#
```

4. The CSV poll log and console output from a chaos_test.py run that passed its SLO, for at least one tier.

a.

```
root@5ec58bd340ab:/testing# python3 chaos_test.py --tier uploader
Target: uploader tier at http://itmo-444-544-app-alb-995129104.us-east-2.elb.amazonaws.com:8080/
Confirming the endpoint is healthy before we start...
Terminating i-08a8a91acd17d654c via the Auto Scaling Group...
Waiting for the endpoint to go down...
Waiting for the endpoint to come back up...
Poll log written to chaos_test_uploader_7075.csv
Recovery time: 69.5s (SLO: 180s)
PASS: recovery time met the SLO.
root@5ec58bd340ab:/testing#
```

b.

```
root@5ec58bd340ab:/testing# cat chaos_test_uploader_7075.csv
elapsed_seconds,is_up
0.0,True
2.12,False
0.0,False
2.12,False
4.27,False
6.41,False
8.54,False
10.69,False
12.84,False
15.0,False
20.07,False
22.22,False
24.37,False
26.5,False
28.63,False
30.78,False
32.9,False
35.03,False
37.21,False
39.33,False
41.46,False
43.61,False
45.74,False
47.89,False
50.03,False
52.19,False
54.35,False
56.5,False
58.61,False
60.73,False
62.86,False
65.04,False
67.18,False
69.51,True
root@5ec58bd340ab:/testing#
```

5. The CSV poll log and console output from a run that failed a tightened SLO.

a.

```
root@5ec58bd340ab:/testing# cd /testing
python3 chaos_test.py --tier uploader --slo-seconds 30
Target: uploader tier at http://itmo-444-544-app-alb-995129104.us-east-2.elb.amazonaws.com:8080/
Confirming the endpoint is healthy before we start...
Terminating i-080a06f69f111975a via the Auto Scaling Group...
Waiting for the endpoint to go down...
Waiting for the endpoint to come back up...
Poll log written to chaos_test_uploader_7335.csv
Recovery time: 67.4s (SLO: 30s)
FAIL: recovery time exceeded the SLO.
root@5ec58bd340ab:/testing#
```

b.

```
root@5ec58bd340ab:/testing# cat chaos_test_uploader_7335.csv
elapsed_seconds,is_up
0.0,True
2.12,False
0.0,False
2.12,False
4.28,False
6.42,False
8.59,False
10.79,False
15.89,False
18.05,False
20.19,False
22.34,False
24.49,False
26.63,False
28.78,False
30.91,False
33.07,False
35.22,False
37.37,False
39.55,False
41.67,False
43.82,False
45.93,False
48.09,False
50.23,False
52.37,False
54.53,False
56.66,False
58.8,False
60.92,False
63.06,False
65.22,False
67.36,True
root@5ec58bd340ab:/testing#
```

6. One paragraph explaining, using your own recovery-time numbers, what dominates the recovery window (health-check timing vs. boot/ npm install time vs. something else).
    a. In my two tests, one run passed an SLO of 180 seconds whereas the second run failed an SLO of 30 seconds. The outage became visible about two to three seconds after I terminated the instance, but the endpoint then stayed down for roughly 67 to 70 seconds before it recovered. A great portion of that recovery window would have been the replacement instance being launched by the Auto Scaling Group. A 10-second interval per instance across two instances would mean that the target group needs roughly 20 seconds of passing checks before the ALB sends traffic to the new instance. Since that detection time is small and fixed by my configuration, boot and install time

dominates the recovery window, hence why instant detection could not have met a 30-second SLO.

7. One paragraph explaining what changed in create_app_stack.sh /delete_app_stack.sh between the previous lab and this one — specifically, what replaced the old run-instances call and why.
    a. The create_app_stack.sh file before would create one uploader and one viewer instance before calling them into an output table. The new code changes this by making the script rely upon the Autoscaling Group (ASG) and Auto-load balancer (ALB) groups to create the instances, so there is no need to manually call the created instances afterwards into a table. This change was necessary because this ties the lifespan of the instance with the ASG, thus simplifying the deletion process in the deletion script. The delete_app_stack.sh file checks for the ASG to delete and removes the ALB security group, which helps to tidy up any leftover groups that can reopen instances.
8. Screenshot of delete_app_stack.sh completing with no errors.
    a.

```
root@5ec58bd340ab:/# ./delete_app_stack.sh 2>&1 | tee delete.log
== Deleting Auto Scaling Groups (this also terminates their instances) ==
Waiting for both Auto Scaling Groups to be gone...
itmo-444-544-uploader-asg deleted.
itmo-444-544-viewer-asg deleted.
== Deleting the load balancer ==
== Deleting target groups ==
== Restoring the app security group to its previous-lab state ==
== Deleting the ALB security group ==
== Looking for stray standalone instances ==
No stray instances found.
== Deleting launch templates ==
{
    "LaunchTemplate": {
        "LaunchTemplateId": "lt-017f3f296c3cdfe0b",
        "LaunchTemplateName": "itmo-444-544-uploader-lt",
        "CreateTime": "2026-10-13T20:00:19+00:00",
        "CreatedBy": "arn:aws:iam::417873488887:user/henry-tran-lab-user",
        "DefaultVersionNumber": 1,
        "LatestVersionNumber": 4,
        "Operator": {
            "Managed": false
        }
    }
}
{
    "LaunchTemplate": {
        "LaunchTemplateId": "lt-02f592167196101a9",
        "LaunchTemplateName": "itmo-444-544-viewer-lt",
        "CreateTime": "2026-10-13T20:00:22+00:00",
        "CreatedBy": "arn:aws:iam::417873488887:user/henry-tran-lab-user",
        "DefaultVersionNumber": 1,
        "LatestVersionNumber": 4,
        "Operator": {
            "Managed": false
        }
    }
}
== Removing IAM instance profiles and roles ==
== Emptying and deleting S3 bucket ==
delete: s3://itmo-444-544-hvtran21-fileshare/shared.txt
Cleanup complete.
root@5ec58bd340ab:/#
```